

3D Personal Portfolio Website

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I. INTRODUCTION

In the dynamic landscape of modern web design, static web pages have given way to interactive experiences that captivate users. The integration of 3D graphics and animations has become increasingly vital in creating visually appealing and engaging web content. As web designers and developers strive to push the boundaries of traditional web design, the adoption of powerful tools becomes essential. One such tool that empowers the creation of stunning 3D visualizations directly within web browsers is Three.js.

A cross-browser JavaScript framework and API called Three.js makes use of WebGL, a low-level graphics API created especially for the web. Three.js provides a platform-independent method for creating intricate 3D animations as a fundamental component of a website by utilizing the JavaScript language.

The motivation behind choosing Three.js lies in its transformative potential and relevance in the current web design landscape. As websites increasingly seek to captivate and engage users, the use of 3D graphics and animations has become a necessity. Three.js provides a powerful and accessible framework for achieving this goal, enabling designers and developers to create immersive and visually striking experiences.

II. RELATED/EXISTING WORK

In the field of 3D personal websites using Three.js, several notable works have emerged. The following examples highlight the achievements and advancements made by designers and developers in this domain.

One example of a 3D personal website created using Three.js is the portfolio of designer and developer, Bruno Arizio [1]. Bruno Arizio's website features a 3D model of his head that follows the user's mouse movements. This creates an engaging and interactive experience for the user. Additionally, the website includes a 3D model of Bruno's workspace that the user can explore.

Another example is the portfolio of designer and developer, Michael Bromley [2]. However, Bromley's website takes things a step further by incorporating a 3D model of his entire body. This creates a more immersive experience for the user and allows them to interact with Bromley in a more personal way.

A third example is the portfolio of designer and developer, Tobias van Schneider [3]. However, what sets Tobias's website apart is the use of sound. The website includes ambient sound effects that create a more immersive experience for the user.

This demonstrates the potential of Three.js for creating not just visually stunning websites, but also websites that engage the user's other senses.

These examples demonstrate the potential of Three.js for creating engaging and interactive 3D personal websites.

III. PROPOSED METHODOLOGY

The proposed methodology focuses on utilizing Three.js to develop 3D personal websites. The goal is to create an immersive and visually appealing web experience that showcases the individual's skills, activities, and personal projects. A 3D portfolio website using Three.js can have the following features:

- 1) **3D Object Showcasing:** Display 3D models of personal projects, designs, or artworks for an interactive and immersive viewing experience.
- 2) **User Interaction:** Allow users to interact with the 3D models using mouse controls, touch gestures, or keyboard inputs. Enable rotation, zooming, and panning of the 3D objects.
- 3) **Animations and Transitions:** Implement dynamic animations and smooth transitions to enhance visual appeal. Create movement or transformation animations for 3D objects or scene transitions.
- 4) **Responsive Design:** Ensure the website adapts to different screen sizes and devices for a seamless user experience on desktop, tablets, and mobile devices.
- 5) **Customization and Personalization:** Enable users to customize the appearance of 3D models, such as changing colors, textures, or materials, to reflect personal preferences or branding.
- 6) **Information Display:** Provide additional information or descriptions about the 3D models or projects, including project details, techniques used, or relevant links.

To implement these features in our 3D portfolio website using Three.js, we have to utilize the functionalities provided by the Three.js library. Three.js offers APIs, controls, and utilities for loading and displaying 3D models, handling user interactions, creating animations, and implementing responsive design.

REFERENCES

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